INTERNET TECHNOLOGIES

QUESTION BANK

UNIT – 1

Short Answer Questions:

1. Define Internet, LAN, MAN, WAN, GIANT WAN, WEB, DNS, Web Browser, Search Engine, WWW, Domain.
2. Differentiate between wired and wireless networks.
3. What is PTP (point – to – point) network? Give example.
4. What is Multi-access / Multi point network? Give example.
5. What is peer to peer network?
6. Mention any two types of P2P network?
7. What is Client / Server network?
8. Give the usage of TCP / IP in Internet.
9. Role of routers in Internet.
10. Define Cyber ethics and Cyber law.
11. What is Network Access Points (NAP's) or Internet Exchange Points(IXP's)?
12. Explain the parts of an URL.
13. **What is Internet Backbone?**

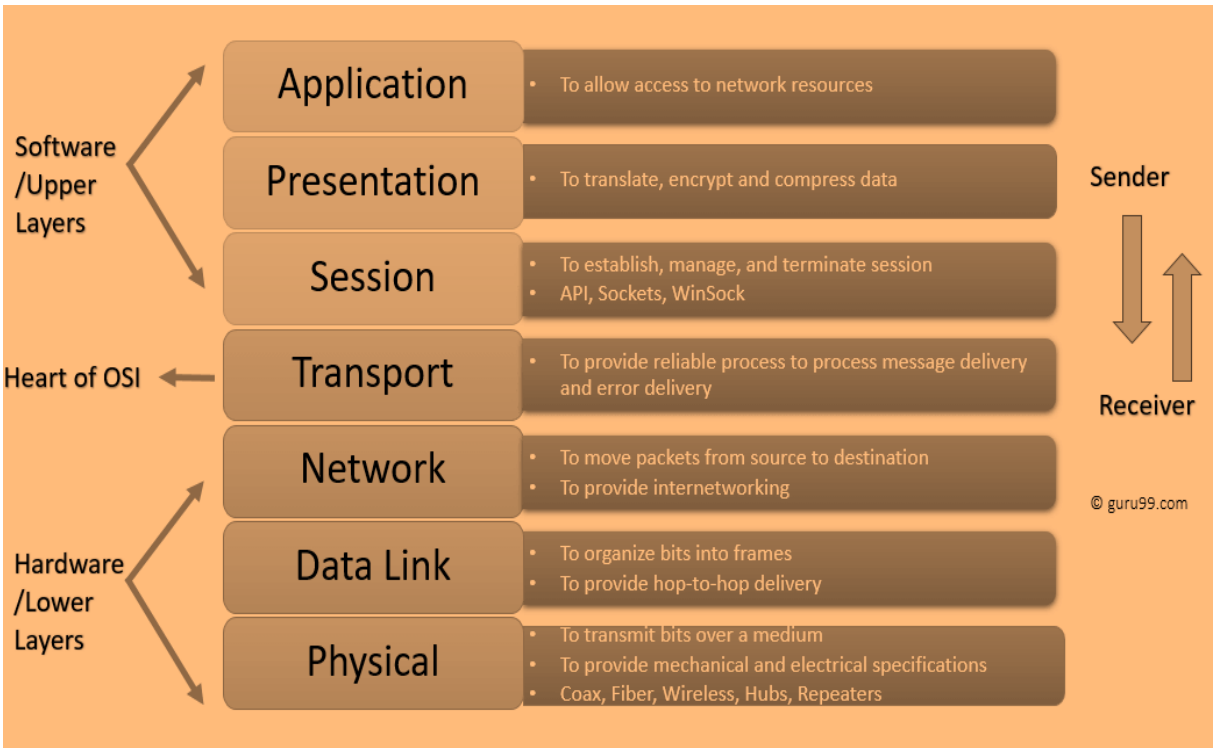
    Ans: The Internet backbone may be defined by the principal data routes between large, strategically interconnected computer networks and core routers of the Internet.
14. **What is nuts and bolts view of internet?**

    Ans: The basic hardware and software components that make up the Internet.
15. Define Internet Services, Email, IRC (Internet Relay Chat), Remote Login, Telnet, Usenet.
16. What is Web Page? Give an example.
17. What is Web Site? Give an example.
18. Role of Web Server in Internet.
19. What are Web Applications? Give an example.
20. What is Web Service? Give an example.

Long Answer Questions:

1. Explain the evolution of the Internet.
2. Explain the different types of network connectivity.
3. Explain different ways of communicating over Internet.
4. How internet works? Explain with examples.
5. **Explain OSI reference model.**

   Ans: OSI stands for Open Systems Interconnection. It has been developed by ISO – 'International Organization for Standardization', in the year 1984. It is a 7-layer architecture with each layer having specific functionality to perform. All these 7 layers work collaboratively to transmit the data from one person to another across the globe.



Layer 1- Physical Layer: The lowest layer of the OSI reference model is the physical layer. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of bits. It is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together.

Layer 2- Data Link Layer (DLL): The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to transmit it to the Host using its MAC address.

Layer 3- Network Layer: The network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender & receiver's IP addresses are placed in the header by the network layer.

Layer 4- Transport Layer: The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as Segments. It is responsible for the End to End Delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found.

Layer 5- Session Layer: This layer is responsible for the establishment of connection, maintenance of sessions, and authentication, and also ensures security.

Layer 6- Presentation Layer: The presentation layer is also called the Translation layer. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network.

Layer 7- Application Layer: At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data, which has to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user.

6. Differentiate between LAN, MAN and WAN.
7. Explain different types of internet access.
8. Explain internet service providers (ISP).
9. Explain any five internet organisations.
10. What are the basic functions of a web browser? Name any four web browsers.
11. Explain the basic building blocks of the internet.
12. What is cyber ethics? Write the important principles of Cyber ethics.
13. Differentiate between client server architecture and peer to peer architecture.
14. Explain the classification of Internet Services.
15. What is simple mail transfer protocol (SMTP)? Explain.
16. What is post office protocol (POP)? Explain.
17. Explain internet message access protocol (IMAP).
18. Explain FTP with advantages and disadvantages.
19. What is VoIP? Explain the components of VoIP.
20. Write a short note on: a) Telnet b) Usenet.

21. Explain working of an email with an example.
22. Explain protocols in email.
23. Explain types of connections and modes in FTP.
24. Explain core Technologies of web / web terminologies.
25. Explain web architecture.
26. Write a short note on challenges in world wide web.
27. Differentiate between two tier and three tier web architecture.

S.NO	Two-Tier Database Architecture	Three-Tier Database Architecture
1	It is a Client-Server Architecture.	It is a Web-based application.
2	In two-tier, the application logic is either buried inside the user interface on the client or within the database on the server (or both).	In three-tier, the application logic or process resides in the middle-tier, it is separated from the data and the user interface.
3	Two-tier architecture consists of two layers : Client Tier and Database (Data Tier).	Three-tier architecture consists of three layers : Client Layer, Business Layer and Data Layer.
4	It is easy to build and maintain.	It is complex to build and maintain.
5	Two-tier architecture runs slower.	Three-tier architecture runs faster.
6	It is less secured as client can communicate with database directly.	It is secured as client is not allowed to communicate with database directly.
7	It results in performance loss whenever the users increase rapidly.	It results in performance loss whenever the system is run on Internet but gives more performance than two-tier architecture.
8	Example – Contact Management System created using MS-Access or Railway Reservation System, etc.	Example – Designing registration form which contains text box, label, button or a large website on the Internet, etc.

UNIT – 2

Short Answer Questions:

1. Define HTTP, HTTPS, Caching, Handshaking, HTTP Cache, Round Trip Time (RTT), Cookies, Semantic Web, Big Data, Web IR, Google PageRank.
2. Mention the parts of HTTP request message.
3. Mention the parts of HTTP response message.
4. Explain any two request methods of HTTP request message.
5. What is persistent and non - persistent HTTP connection?
6. Mention any two characteristics of big data.
7. Give any two examples of big data mention the types of web search tools.
8. What are crawlers? What is the role of crawlers in web IR?

Long Answer Questions:

1. Write the characteristics of HTTP.
2. Explain persistent and non - persistent HTTP connection.
3. Explain URL in detail.
4. Differentiate HTTP Vs HTTPS.
5. Write a short note on cookies explain the working of HTTP.
6. Explain the different versions of HTTP.
7. Explain HTTP request message and response message in detail.
8. Explain hypertext transfer protocol state retention in detail with examples.
9. Explain the evolution of Web.
10. Explain the generations of Web.
11. Give the differences between Web 1.0, Web 2.0, and Web 3.0
12. Discuss in detail about Big Data with advantages and disadvantages.
13. Explain Web IR Tools.
14. Explain Web IR Architecture.
15. Explain the types of Web IR models.
16. Discuss briefly about Google PageRank.
17. Write a note on Google PageRank Algorithm.
18. Write a short note on: a) Indexing Process b) Crawlers.

UNIT - 3

Short Answer Questions:

1. Define Web Browser, Web Server, Web Development, Front End & Back End Development, Full Stack Web Development, Scripting Language, API's, GitHub, Web Hosting. HTML, hyperlink, href attribute, Form, Frame, CSS, JavaScript, Bootstrap Framework, AngularJS Framework.
2. Mention any two Client Side scripting language.
3. Mention any two Server Side scripting language.
4. Give any two examples of Scripting language.
5. How HTML works?
6. Write a simple HTML code to display "Hello BCA".
7. Give the types of heading tags.
8. Give the difference between ordered and unordered lists.
9. Give the syntax to create a table in HTML.
10. How can you specify the width and height of an image in HTML?
11. Give the usage of title tag.
12. What is the purpose of label tag?
13. Explain <th>, <tr> and <td> tags.
14. What is rowspan and colspan?
15. Give the usage of CSS.
16. What is Ajax?
17. What are Server Side Technologies?
18. Mention any two Server Side Technologies.
19. What is PHP?
20. How to create variable in PHP? Give an example.
21. How to create constant in PHP? Give an example.
22. Mention any two benefits of NodeJS.
23. Mention any two benefits of PHP.
24. Give the syntax of JavaScript comments.
25. Give the usage of <!DOCTYPE html>.
26. Give the usage of document.write() .
27. What is the usage of **let** keyword?

Long Answer Questions:

1. What is Web Development? Explain the classification of Web Development.
2. **Explain Web Development Process.**

   Ans: The web development process involves a series of steps to create, design, develop, and deploy a website or web application.

a. **Planning:** In the stage, web development team identifies the goals and requirements of the project, and create a plan for how to achieve them.
   Example: if the goal is to create an E-commerce website, the team might identify the required features such as Shopping Cart payment gateway and user account management.

b. **Design:** In the stage, the web development team create official design for the website, including the layout colour scheme and typography.
   Example: the team might create wire frames of the website using design tools such as Adobe XD or sketch.

c. **Development:** In the stage, the web development team rights the code for the website, using programming languages or Technology such as HTML, CSS, and JavaScript.
   Example: The team might create the front end of the website using HTML and CSS and the back end using a server side language such as PHP or Python.

d. **Testing:** In the stage, the web development team tests the website to ensure that it functions correctly and meets the requirements.
   Example: The team might test the website functionality, performance and security using tools such as Selenium or Jmeter.

e. **Deployment:** In this stage the web development team deploys the website to a web server, making it available to the public making it available to the public.
   Example: The team might deploy the website to a cloud hosting service such as Amazon Web Services or Microsoft Azure.

3. Give the differences between programming language and scripting language.
4. Explain the types of scripting language.
5. Differentiate Client Side and Server Side Scripting.
6. Explain MVC Architecture for Web Application Development with advantages and disadvantages.

7. Explain the features or characteristics of HTML.
8. Explain basic HTML tags with examples.
9. Describe the syntax and attributes of the HTML image tag, and explain how it can be used to display images on web page.
10. Explain CSS with advantages and disadvantages.
11. Explain the types of style sheets or levels of CSS with an example.
12. What is JavaScript? Write the features of JavaScript.
13. Explain JavaScript Operators or Control Structures.
14. Explain JavaScript functions with a suitable example.
15. Explain the JavaScript message boxes (alert(), confirm() and prompt()) with a suitable example.
16. Explain the Bootstrap framework.
17. Explain AngularJS framework.
18. Explain the Model-View-Controller Architecture of AngularJS.
19. Explain the structure of HMTL with an example.
20. Explain Server Side Technologies and Server Side Scripting with examples.
21. Explain how to create a sample page in PHP.
22. Explain functions in PHP with examples.
23. How Node.js works? Mention the advantages and disadvantages of Node.js.
24. Explain Arrays in PHP.
25. **Program to find the factorial of a number using JavaScript**

```
<!DOCTYPE html>
<html>
<head>
</head>
<body style = "text-align: center; font-size: 20px;">
Enter a number: <input id = "num">
<br><br>
<button onclick = "fact()"> Factorial </button>
<p id = "res"></p>
<script>
function fact(){
```

```
var i, num, f;
f = 1;
num = document.getElementById("num").value;
for(i = 1; i <= num; i++)
{
f = f * i;
}
i = i - 1;
document.getElementById("res").innerHTML = "The factorial of the
number " + i + " is: " + f ;
}
</script>
</body>
</html>
```

UNIT - 4

Short Answer Questions:

1. What is Web Application framework?
2. What is Django?
3. What is Ruby and Ruby on Rails?
4. Define Web Databases, Relational Database, NoSQL / Non-Relational Database, Document Database, Column Store, Key-Value, Graph Database.
5. Define Contextual Retrieval on the Web.
6. What is Contextual Information Retrieval?
7. Define Web Mining, Data Mining, Video Mining, Audio Mining.

Long Answer Questions:

1. Explain the Model-View-Template Architecture of Django.
2. Explain the Model-View-Controller Architecture of Ruby on Rails.
3. Explain the categories of Web Databases with examples.
4. Write a short note on a) MYSQL b)MongoDB c)Neo4j
5. Compare relational (SQL) and non-relational databases (NoSQL).
6. Explain Web Mining Classification.
7. Explain Sentiment Analysis Approaches.
8. Write a short note on popular web databases.
9. Explain the categories of NoSQL.
10. Define Distributed Database. Explain the types of distributed database.
11. Discuss about research trends on the web.

Note: Apart from these questions, you must be prepared with Practical Programs for theory examination.